EX: 14
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IMPLEMENTING PACKET SNIFFING USING RAW SOCKETS

AIM:

To implement packet sniffing using Raw socket Python.

ALGORITHM:

1) Create a raw socket
2) Bind the socket to network interface
3) Receive packets continously.
4) Read packet header.
5) Display source and distination IP address
6) Display the protocol used.
7) Continue until user stop the process.

INPUT:

NETwork packets received through the network interface.

CODE:

```
import socket.
S= socket.socket(socket.AF_INET, SOCKET.SOCK_RAW, socket
S.bind(("0.0.0.0", 0))
S.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
print("packet sniffer started...")

While True:
    packet, addr = S.recvfrom(65565)
    ip header = packet[0:20]
    iph = struct.unpack("!BBHHHBBH4s4s", header)

str_IP = socket.inet_ntoa(iph[8])
des_IP = socket.inet_ntoa(iph[9])
```

```
print ("source ip :", saddr)
print ("Destination ip :", daddr)
print ("packet count :", len(packet))
```

OUTPUT:

packet sniffer started....
Source ip 192.168.1.10
Destination ip 192.168.1.1
packet count : 79

RESULT:

Thus, a raw socket program was successfully implement to capture and display network packets